

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangam, Machhe, Belagavi-590018



Scheme of Teaching and Examinations 2025

Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS) (Effective from the academic year 2025-26)

B.E. in Robotics & Automation Scheme of Teaching and Examinations 2025
Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS) (Effective from the academic year 2025-26)

III SEMESTER

S l · N o	Course	Course Code	CourseTitle	Teaching Department and Question Paper Setting Board (PSB)	Teaching Hours/Week				#TH/S	Examination				Credits
					CI		LI	TW & SL		Duration in hours	CIE Marks	SEE Marks	Total Marks	
1	ASC/PCC	1BRA301	Discrete Mathematics and Probability Theory	TD /PSB: Mathematics/ Specific Dept.	42	28	00	50	120	3	50	50	100	4
2	IPCC	1BRA302	Data Structures and Algorithms	TD/PSB: CSE/RA	42	00	28	50	120	3	50	50	100	4
3	PCC	1BRA303	Mechanics of Materials	TD/PSB: ME/RA	42	28	00	50	120	3	50	50	100	4
4	PCC	1BRA304	Analog and Digital Electronics (ADEC)	TD/PSP: ECE/RA	42	00	00	48	90	3	50	50	100	3
5	PCC	1BRA305	Fundamentals of Industrial Robots	TD/PSB: ME/RA	42	00	00	48	90	3	50	50	100	3
6	PCCL	1BRAL306	Computer Aided Modeling Lab	TD/PSB: ME/RA	0	0	28	2	30	2	50	50	100	1
7	AEC	1BRAL307A	Analog and Digital Electronics Circuits Lab	TD/PSB: ECE/RAI	0	0	28	2	30	2	50	50	100	1
8	SDC	1BSC308	Community Project (Project-Based Learning/Societal Project)	Any Department/Respective Engineering Department	0	0	0	30	30	2	50	50	100	1
9	NCMC	1BNSS309	National Service Scheme (NSS)	NSS Coordinator	0	0	0	28	28	--	100	---	100	PP
		1BPE309	Physical Education (PE) (Sports and Athletics)	Physical Education Director										
		1BYOG309	Yoga	Yoga Teacher										
		1BMUK309	Music	Music Teacher										
	Total										500	400	900	21
10	NCMC	1BMATDIP310	Mathematics course for Lateral Entry Students	TD/PSB-Maths Dept	14	0	0	14	28	03	100	---	100	PP
Total									686		600	400	1000	21

#TH/S – Total Hours per Semester, TW & SL – Team Work & Self Learning

ASC- Applied Science Course, PCC- Professional Core Course, IPCC-Integrated Professional Core Course, PCCL- Professional Core Course Laboratory, AEC-Ability Enhancement Course, SDC- Skill Development Course, NCMC- Non-Credit Mandatory Course, SL-Term-work and Self Study hours, CIE-Continuous Internal

The course **1BXXL307 – Ability Enhancement Course Laboratory** shall be offered either as a single compulsory course or as multiple options under the course codes **1BXXL307x** (where x = A, B, C, D). Each student shall choose **one course** from the available options for study. Since this is a laboratory course, the **minimum strength required** for each course shall be **20 students**. If the number of students opting for a particular course is **less than 20**, the college shall seek permission from the **Registrar, VTU Belagavi**, to offer that course.

Professional Core Course (IPCC): The Integrated Professional Core Course (IPCC) refers to a core theory course that is integrated with practical components of the same subject. Each IPCC carries 4 credits, with Teaching–Learning hours structured as either (3:0:2) or (2:2:2). The theory component of the IPCC shall be evaluated through both Continuous Internal Evaluation (CIE) and Semester End Examination (SEE). The practical component shall be assessed exclusively through CIE, with no separate SEE. However, questions derived from the practical component may be included in the SEE question paper to ensure comprehensive evaluation.

National Service Scheme / Physical Education / Yoga (NSS / PE / YOG): All students are required to register for any one of the following courses—National Service Scheme (NSS), Physical Education (PE) (Sports and Athletics), or Yoga (YOG)—with the respective course coordinator during the first week of the third, semesters.

- Colleges shall submit Continuous Internal Evaluation (CIE) marks each semester based on the activities completed by students under the selected course.
- Students may opt for different activities/courses across semesters. For instance, a student participating in PE during the 3rd semester may choose NSS in the 4th semester or Yoga.
- This flexible approach aligns with the student-centric vision of the National Education Policy (NEP) 2022, and facilitates equitable distribution of NSS/PE/Yoga-related responsibilities across departments.
- Activities shall be conducted over two semesters (III & IV), and successful completion of the registered course along with the required CIE score is mandatory for the award of the degree.
- Institutions must ensure that events are appropriately scheduled and reflected in the semester-wise calendar for NSS, PE, and Yoga activities.

These courses shall not be considered for vertical progression, nor shall they be included in the calculation of SGPA or CGPA. However, completion is compulsory for degree eligibility.

Societal project work: Students shall undertake Societal Project Work to address real-world social challenges and contribute meaningfully to community well-being. They shall engage with domains such as health, education, sustainability, digital inclusion, and rural development through experiential learning. Students shall apply their domain-specific knowledge to design and implement solutions that respond effectively to identified societal needs. By interacting directly with communities and stakeholders, they shall develop practical skills, empathy, and ethical awareness.

A **Non-Credit Mandatory Course (NCMC)** shall constitute an integral component of the academic framework, aimed at enhancing students' knowledge, skills, and awareness beyond the core curriculum. These courses shall not carry academic credits nor be considered in the computation of CGPA; however, successful completion of the NCMC shall be compulsory for fulfilling the requirements of the academic program. Each student shall register for the designated NCMC in the prescribed semester as notified by the University/Institution. NCMCs shall not be considered for vertical progression. However, qualifying in the NCMC is mandatory for the award of the degree. A student who fails to qualify in the prescribed NCMC shall not be eligible for the conferment of the degree.

Credit Definition:

1-hour Lecture (**L**) per week=**1Credit**

2-hours Tutorial (**T**) per week=**1Credit**

04-Credit courses are designed for 50 hours of Teaching-Learning sessions

04-Credit (IC) courses are designed for 40 hours' theory and 10-12 hours of practical sessions

03-Credit courses are designed for 40 hours of Teaching-Learning Session

2-hours Practical / Drawing (P) per week= 1Credit	02- Credit courses are designed for 25 hours of Teaching-Learning Session 01-Credit courses are designed for 12 hours of Teaching-Learning sessions
AICTE Activity Points Requirement for BE/B.Tech. Programmes (Lateral Entry Students) As per AICTE guidelines (refer to Chapter 6 – AICTE Activity Point Program, Model Internship Guidelines), in addition to academic requirements, students must earn a specified number of Activity Points to be eligible for the award of the degree. The points to be earned is: 1. Regular students admitted to a 4-year degree program must earn 100 Activity Points. 2. Lateral entry students (joining from the second year) must earn 75 Activity Points. 3. Students transferred from other universities directly into the fifth semester must earn 50 Activity Points from the date of entry into VTU. These Activity Points are non-credit and will not be considered for the SGPA/CGPA or be used for vertical progression. However, earning Activity Points is mandatory for the award of the degree, and the points earned will be reflected on the eighth semester Grade Card. If a student completes all the semesters (eight or six) at the end of the programme but fails to earn the required Activity Points, the eighth-semester Grade Card will be withheld until the requirement is fulfilled. Also, the degree will be awarded only after the Grade Card has been released. The hours spent earning the activity points will not be counted for regular attendance requirements. Students can accumulate these points at any time during their program period, including weekends, holidays, and vacations, starting from the year of admission, provided they meet the minimum hours of engagement prescribed for each activity by AICTE.	
Ability Enhancement Course Laboratory (May be one course or	
1BRAL307A	Analog and Digital Electronic Circuits lab

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IV SEMESTER															
Sl. No	Course and Course Code		Course Title	Teaching Department (TD)and Question Paper Setting Board (PSB)	Teaching Hours/Week					Examination				CREDITS	
					CI		LI	TW & SL	# TW/S	Duration in Hrs/week	CIE Marks	SEE Marks	Total Marks		
					L	T	P								
1	ASC/PCC	1BRA401	Robot Kinematics and Dynamics	TD/PSB: ME/RA	28	28	00	34	90	3	50	50	100	3	
2	IPCC	1BRA402	Object Oriented Programming	TD/PSB: SE/RA	42	00	28	50	120	3	50	50	100	4	
3	PCC	1BRA403	Control Systems	TD/PSB: ME/RA/ECE/EE	42	28	00	50	120	3	50	50	100	4	
4	PCC	1BRA404	Sensors and Actuators	TD/PSB: ME/RA/ECE/EE	28	28	00	34	90	3	50	50	100	3	
5	PCCL	1BRAL405	Sensors and Actuators Laboratory	TD/PSB: ME/RA/ECE/EE	00	00	28	02	30	2	50	50	100	1	
6	AEC	1BRAL406A	Industrial Robot Laboratory	TD/PSB: ME/RA	00	00	28	02	30	2	50	50	100	1	
7	BSC	1BRA407	Biology for Engineers	TD/PSB: Biology/Chem	28	00	00	32	60	3	50	50	100	2	
8	SDC	1BEP408	Environmental Science Project (Interdisciplinary Project Based Learning)	TD/PSB: Any Department	00	00	00	30	30	3	50	50	100	1	

9	PCC/PEC	1BRA409	Design of Robotic Components	TD/PSB: ME/RA	28	28	00	34	90	3	50	50	100	3
9	NCMC	1BNSK409	National Service Scheme (NSS)	NSS coordinator	0	0	0	28	28	--	100	---	100	PP
		1BPEK409	Physical Education (PE) (Sports and Athletics)	Physical Education Director										
		1BYOK409	Yoga	Yoga Teacher										
		1BMUS409	Music	Music Teacher										
Total									688		550	450	1000	
10	NCMC	1BMATDIP410	Mathematics for Lateral Entry Students	Maths Dept	14	0	0	14	28	--	100	--	100	PP
Total									716		650	450	1100	22

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to qualify in the prescribed NCMC shall not be eligible for the conferment of the degree.

An **Environmental Science project** for engineering students shall aim to integrate ecological awareness with technical innovation. The project shall be either interdisciplinary—drawing from multiple branches of engineering—or domain-specific, focusing on a particular field. It shall encourage students to apply sustainable practices, analyze environmental impact, and propose viable solutions. The chosen topic shall reflect current environmental challenges and align with academic objectives. Students shall document their findings systematically and present actionable outcomes. The project shall foster responsible engineering aligned with environmental stewardship.

The course **1BXXL406 – Ability Enhancement Course Laboratory** shall be offered either as a single compulsory course or as multiple options under the course codes **1BXXL406x** (where $x = A, B, C, D$). Each student shall choose **one course** from the available options for study. Since this is a laboratory course, the **minimum strength required** for each course shall be **20 students**. If the number of students opting for a particular course is **less than 20**, the college shall seek permission from the **Registrar, VTU Belagavi**,

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Ability Enhancement Course Laboratory			
1BRAL406A	Industrial Robot Lab	1BxxL406C	